TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

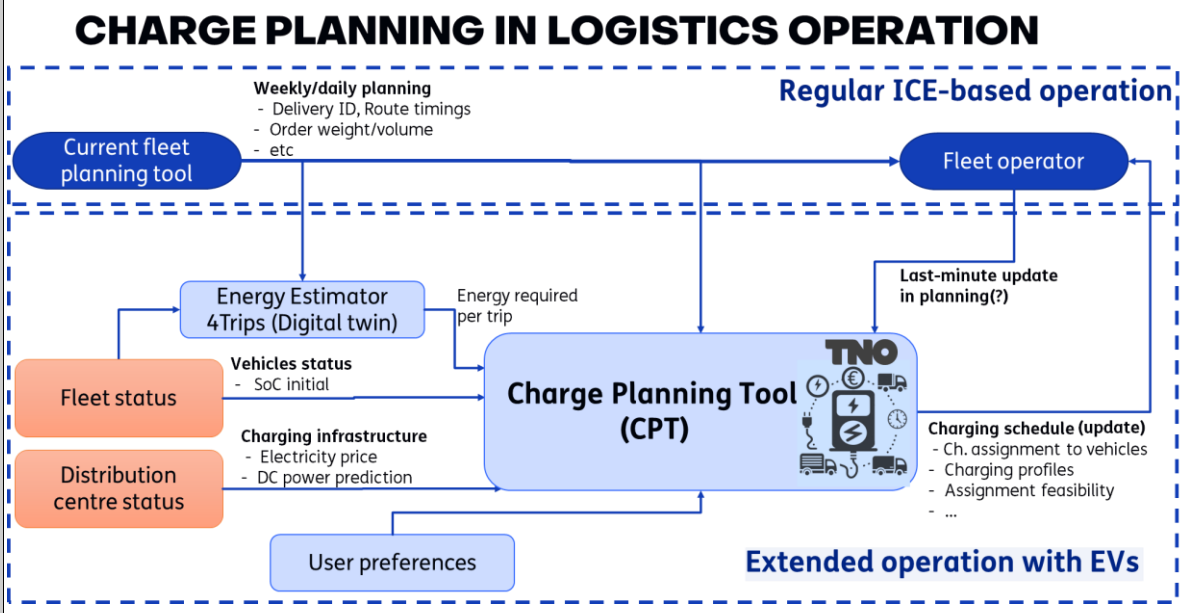
The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Subhajeet Rath	Email	subhajeet.rath@tno.nl
Institute Name	Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk Onderzoek (TNO), Powertrains Department			
Address	Automotive Campus 30, 5708 JZ Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Life-Aware Smart Charging Platform with Charge Planning, Battery Digital Twins, and Charger-Based Battery Diagnostics for Grid-Constrained Depots			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-generation Energy (√), Bio-health ( ), Aerospace ( ), Quantum ( ), Advanced Robotics & Manufacturing ( ), AI ( ), Cybersecurity ( ), Hydrogen ( ), Semiconductor materials ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	Internet SW	Batteries	Secondary Batteries
Concept of Proposed Technology	**Core concept** An interoperable software stack that couples three capabilities to reduce charging cost and grid impact while improving battery lifetime for large EV fleets and charging depots: • Charge Planning Tool (CPT): multi-objective charge scheduling that accounts for depot grid limits, charger constraints, time-varying tariffs, and battery degradation costs. • Battery Digital Twin (DT): adaptive battery aging and charge-profile prediction models that self-calibrate using operational data and provide forecasting to the CPT.			

	• Charger-based diagnostics: in-service battery health checks executed through (bi-directional) chargers during regular charging to estimate state-of-health and detect anomalies early. Key technical enablers: open charging protocols (OCPP for EVSE-backend, ISO 15118 for EV-EVSE communication, OCPI for roaming) and a secure data pipeline for time-series telemetry, events, and model outputs.  Relevant TNO background: CPT and fleet charging scheduling research by Wilkins and co-authors; digital-twin based charge profile prediction; and VITALISE battery diagnostics extension with the CheckUp tool. Citations: (1)-(5), (10)-(15), (17).
Rationale for Joint R&D Project	**Need and rationale** Large-scale depot charging is increasingly constrained by local grid capacity, dynamic electricity prices, and the need to coordinate with on-site renewables and storage. Energy hub concepts aim to match local supply and demand in time and location to relieve the grid, which requires advanced planning and control software. For fleet operators, uncertainty in routes, arrival times, and energy demand makes static charging plans brittle. TNO has demonstrated fleet charge planning methods that explicitly handle operational constraints and can incorporate robustness and degradation-aware objectives. Battery lifetime is a major component of total cost of ownership. Recent TNO and academic work (including Wilkins as co-author) integrates degradation and remaining useful life considerations into fleet charge scheduling. Charger-based diagnostics can reduce downtime and enable predictive maintenance by assessing battery health during normal charging operations, as demonstrated in TNO's VITALISE battery diagnostics extension and CheckUp tool concept.

	A joint R&D project is required to integrate these methods into an operational smart charging platform, validate at scale with real-world charger and fleet data, and demonstrate grid services such as peak shaving and demand response while maintaining operational reliability. Citations: (2), (4), (6), (10)-(11), (17).
Key Words	1. Smart charging 2. Fleet charge planning 3. Battery digital twin 4. Charger-based diagnostics 5. Energy hubs and demand response

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	• Develop and validate an integrated charge planning, digital twin, and diagnostics toolchain suitable for deployment in a large-scale smart charging platform (backend). • Implement degradation-aware and robustness-aware scheduling for mixed fleets under grid capacity limits and variable tariffs. • Develop adaptive battery digital twins for (a) battery aging prediction and (b) charge profile prediction, with online calibration using operational telemetry. • Develop charger-based diagnostic routines and health estimation models that can be executed during regular (bi-directional) charging to estimate state-of-health and detect anomalies. • Demonstrate interoperability using open protocols (OCPP 2.0.1, ISO 15118, OCPI) and deliver a reference architecture and APIs for integration. Citations: (1), (3)-(5), (10)-(15), (17).			
R&D Target Specifications	**Target specifications (project targets, to be verified in pilots)** Charge planning performance • Generate feasible schedules for depots with 50-500 vehicles and heterogeneous chargers within 5 minutes for day-ahead planning; rolling re-optimization within 60 seconds upon disruptions. • Reduce peak grid import (kW) by 10%-20% relative to uncontrolled charging while meeting all departure energy requirements (SoC targets). • Support objective functions including electricity cost (time-of-use), demand charges, and battery degradation proxies used in published scheduling work. Digital twin accuracy • Charge profile prediction MAPE below 10% at 1 minute resolution for representative DC charging sessions after adaptation. • Aging prediction capacity-loss error below 5% for representative duty cycles over multi-month horizons using periodic calibration. Diagnostics via charger • State-of-health (capacity) estimation error below 3%-5% compared with reference lab tests for selected packs, using in-service diagnostic routines through chargers. Interoperability and security • OCPP 2.0.1 smart charging support for setpoint control and telemetry; ISO 15118 secure EV-EVSE			

	communication; OCPI roaming interoperability. Citations: (4)-(5), (12)-(15), (17).
R&D Plan	Overall duration: 36 months (example). Work plan can be adapted to TIPA call constraints. **WP1 Requirements and system architecture (M0-M6)** • Define depot and fleet use-cases, constraints, and KPIs. Specify data interfaces: OCPP (EVSE-backend), ISO 15118 (EV-EVSE), OCPI (platform roaming). Define cybersecurity and data governance baseline. **WP2 Charge Planning Tool integration and algorithms (M3-M18)** • Implement multi-objective scheduling (cost, peak shaving, robustness, degradation). Develop day-ahead planning and real-time re-optimization. Benchmark against published CPT algorithms and scale-up scenarios. **WP3 Battery Digital Twin development (M3-M24)** • Develop adaptive DTs for battery aging and charge profile prediction. Integrate DT outputs into CPT optimization. Calibrate and validate using operational data and controlled lab datasets. **WP4 Charger-based diagnostics (M6-M30)** • Develop charger-executed diagnostic routines and health estimation. Validate with lab reference tests at TNO Battery Lab and pilot chargers. Define decision thresholds for maintenance and anomaly alerts. **WP5 Demonstration and validation (M18-M36)** • Deploy in a pilot depot or controlled field environment. Validate cost, grid impact, and battery health KPIs. Document reference architecture, APIs, and deployment guidelines. **WP6 Dissemination and exploitation (M0-M36)** • Publications, workshops, and standardization feedback (OCPP, ISO 15118). Joint exploitation plan and IP strategy. Citations: (6)-(9), (12)-(15), (17).

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**TNO (overseas research institute): expected roles and responsibilities** • Provide the scientific and engineering lead for charge planning, battery digital twin development, and charger-based diagnostics, leveraging published CPT and DT methods (Wilkins and co-authors). • Develop optimization formulations and scalable solvers, including robustness and degradation-aware objectives. Provide reference implementations and validation benchmarks. • Develop DT models and calibration methods; integrate DT outputs into CPT for life-aware scheduling. • Develop charger-based diagnostic concepts and validation protocols aligned with VITALISE and CheckUp work. • Validation and testing at TNO facilities: Battery Lab (cell/module/pack testing) and Powertrains Test Centre with climate chambers and electrical test benches, including 300 kW bi-directional electrical load/supply capability in dedicated test cells. • Provide documentation, training, and dissemination (joint papers, workshops, standardization input). Citations: (1), (4)-(5), (7)-(11), (17).
Expected R&R of Korean SME Partner	**Korean SME partner: expected roles and responsibilities** • Provide platform integration environment for smart charging backend and access to representative operational data streams: charger telemetry, charging sessions, tariffs, and depot constraints. • Implement and test the integration interfaces (OCPP/OCPI/ISO 15118 where applicable) and support deployment in pilot settings. • Co-develop user requirements, KPIs, and acceptance criteria for fleet operators and charging site owners. • Support demonstration activities: select pilot sites, coordinate stakeholder engagement, and operate the platform during the trial period. • Lead commercialization planning in the Korean market and contribute to joint IP and exploitation strategy.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Expected outcomes and impact** Technical outcomes • Operational smart charging algorithms: life-aware, robustness-aware scheduling for grid-constrained depots aligned with published CPT approaches. • Battery digital twin modules: adaptive aging prediction and charge profile prediction integrated into planning workflows. • Charger-based battery diagnostics workflow to estimate state-of-health during charging for predictive maintenance. • Reference architecture and APIs: protocol-compliant integration with OCPP 2.0.1, ISO 15118, and OCPI, including cybersecurity baseline. Impact • Reduced grid impact and improved asset utilization via peak shaving and better coordination of charging with depot operations and energy hubs. • Lower total cost of ownership through reduced electricity costs and battery lifetime extension supported by degradation-aware scheduling research. • Improved reliability and safety through early battery health detection and actionable diagnostics during charging. • Strengthened international collaboration and knowledge transfer through joint publications and standardization feedback. Citations: (2), (4), (6), (10)-(11), (12)-(15), (17).

6. References

(1) Charge Planning Tool for Heavy-Duty Electric Vehicle Fleets (TNO Repository record). Accessed 2025-12-19. https://resolver.tno.nl/uuid:4043c8dc-e109-4441-8b23-0ea3a2fddf68

(2) Optimal charging planning for logistics service providers (TNO Insight). Accessed 2025-12-19. https://www.tno.nl/en/newsroom/insights/2024/12/optimal-charging-planning-logistics/

(3) Rath, S.; Medina, R.; Wilkins, S. Real-time Optimal Charging Strategy for a Fleet of Electric Vehicles Minimizing Battery Degradation. IEEE SeFeT 2023. DOI:10.1109/SeFeT57834.2023.10245996. Accessed 2025-12-19. https://doi.org/10.1109/SeFeT57834.2023.10245996

(4) Geerts, D.; Medina, R.; van Sark, W.; Wilkins, S. Charge Scheduling of Electric Vehicle Fleets: Maximizing Battery Remaining Useful Life Using Machine Learning Models. Batteries 2024, 10(2), 60. DOI:10.3390/batteries10020060. Accessed 2025-12-19.

https://doi.org/10.3390/batteries10020060

(5) Rath, S.; Wilkins, S. Prediction of Electric Vehicle Charge Profile using Battery Digital Twin (TNO Repository record). Accessed 2025-12-19. https://resolver.tno.nl/uuid:a4b05a75-0e61-41a7-8368-f006653ff414

(6) Charging Energy Hubs (TNO project page). Accessed 2025-12-19. https://www.tno.nl/en/sustainable/mobility-logistics/energy-management/charging-energy-hubs/

(7) Sustainable batteries and advanced battery management (TNO topic page). Accessed 2025-12-19. https://www.tno.nl/en/sustainable/mobility-logistics/batteries/

(8) Battery Lab (TNO facility page). Accessed 2025-12-19. https://www.tno.nl/en/technology-science/facilities/battery-lab/

(9) TNO research on powertrains (Powertrains Test Centre and facilities). Accessed 2025-12-19. https://www.tno.nl/en/about-tno/organisation/units/mobility-built-environment/tno-research-powertrains/

(10) VITALISE: Battery diagnostics extension (TNO project page). Accessed 2025-12-19. https://www.tno.nl/en/sustainable/mobility-logistics/batteries/vitalise-battery-diagnostics-extension/

(11) The health of electric bus batteries is now measurable on-site through a charging cycle (TNO News, 12 Apr 2025). Accessed 2025-12-19. https://www.tno.nl/en/newsroom/2025/04/health-electric-bus-batteries/

(12) Open Charge Point Protocol (OCPP) overview (Open Charge Alliance). Accessed 2025-12-19. https://openchargealliance.org/protocols/open-charge-point-protocol/

(13) Open Charge Point Protocol 2.0.1 specification (PDF via US DOT docket attachment). Accessed 2025-12-19. https://downloads.regulations.gov/FHWA-2022-0008-0404/attachment_6.pdf

(14) OCPI 2.2.1 (EV Roaming Foundation) specification (PDF). Accessed 2025-12-19. https://evroaming.org/wp-content/uploads/2024/11/OCPI-2.2.1-d2.pdf

(15) ISO 15118-20:2022 Road vehicles - Vehicle to grid communication interface - Part 20 (ISO standard page). Accessed 2025-12-19. https://www.iso.org/standard/77845.html

(16) Battery Modelling & Battery Management Systems (TNO publication PDF, 2024). Accessed 2025-12-19. https://publications.tno.nl/publication/34642258/wUCI2v/TNO-2024-battery.pdf

(17) Rath, S.; Beckers, A.; Netto, P.; Medina, R.; Wilkins, S. Charge Scheduling for a Fleet of Electric Vehicles using Battery Digital Twin (EVS38 proceedings, 2025). Accessed 2025-12-19. https://evs38-program.org/images/Proceedings/D%20Charging%20Infrastructure%20and%20grid%20integration/197_Charge%20Scheduling%20for%20a%20Fleet%20of%20Electric%20Vehicles%20using%20Battery%20Digital%20Twin.pdf

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization